PRODUCT SCOPE · INPUT TO SPEC 0.3

Minmed Digital Name Cards

A digital name card for every Minmed staff member, managed centrally. Admins create and change every card. Staff sign in only to see their card, add a name or alias to their link and download their QR code. Anyone with a card's link can open it, but nobody can find it on Google.

Based on Technical Specification draft 0.2 (15/10/2026) and your decisions on 16/10/2026. Names, numbers, codes and QR codes in the examples are made up. Updated 16/10/2026.

CONFIRMED you decided it

PROPOSED my recommendation, waiting for your OK (section 11)

TEST try on a test app before the build

1

How it works

1

Admin adds the card

In the admin console, one at a time or from a spreadsheet.

2

Staff member gets an email

Their link and QR code, an Add to Google Wallet button, and steps for Apple Wallet.

3

Staff member can add a name to the link

Optional. They sign in with their Minmed Google account and type a name or alias. They get a new link and a new card email; the old link stops working.

4

Staff member shares it

Sends the link on WhatsApp or email, or shows the QR code from Wallet.

5

Visitor opens the link

Sees the card, then saves the contact, emails, calls or WhatsApps. Unusual traffic gets a CAPTCHA.

6

Details change

The staff member replies to the card email. An admin edits the card, and the same link shows the new details.

7

Staff member leaves

HR tells an admin. On the last day the card is deactivated and soft-deleted, and the staff member can't sign in. The link shows “This card is no longer active.”

Figure 1. The life of a card, from creation to deactivation.

2

What's confirmed

On 16/10/2026 you approved the proposals in this document. Open items are in section 11.

PRODUCT

One central place to manage every staff card. Staff can't change their card details, so there's no approval process. The only thing they can change is the name or alias in their link.

SOURCE
You, 16/10

Cards are public to anyone with the link, but can't be found through Google.

SOURCE
You, 16/10

Each link ends in a random code of 22 letters and numbers, so nobody can guess it: `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w`. Staff can put a name or alias in front of the code: `me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`.

SOURCE
You, 16/10; industry standards

Staff sign in with their Minmed Google account to a My card page at `me.minmed.sg/my-card`. There they can only read their card details, set or change the name or alias in their link, and download their QR code. They keep their link on their phone, or in Apple Wallet or Google Wallet.

SOURCE
You, 16/10

Built on Base44. Phone-first public card. Admins and staff sign in with company Google accounts.

SOURCE
Spec 0.2 §1.1; you, 16/10

THE CARD

Fields: name, designation, Minmed, work email, optional mobile and photo, optional biography, languages and LinkedIn. Company information is a link to minmed.sg; no social links.

SOURCE
Spec 0.2 §3.1; you, 16/10

Actions: Save contact and Email, plus Call and WhatsApp when a mobile is published. No visitor app.

SOURCE
Spec 0.2 §1.1; you, 16/10

Every card shows the Minmed logo. A Super admin sets it once in Settings; it's the same on every card and can't be changed for a single card. The card forms and My card show it as locked.

SOURCE
You, 16/10

On a computer, the card also shows a QR code to move it to a phone.

SOURCE
You, 16/10

No “Active staff profile ✓ Checked” line and no How to verify panel.

SOURCE
You, 16/10

Sharing the link on WhatsApp shows a preview of the card. **AT RISK** Base44 can't make a per-card preview as is. See section 10 and open item 1.

SOURCE
Spec 0.2 §4.4

WALLET

Minmed issues a Google Wallet pass. iPhone staff add their QR code to Apple Wallet themselves with iOS 27's Create a Pass. No Apple Developer Program, fee or D-U-N-S number.

SOURCE
You, 16/10

NAME OR ALIAS IN THE LINK

Optional. Letters, numbers and hyphens, 2 to 30 characters.

SOURCE
You, 16/10

Until an alias is set, the plain link without a name works.

SOURCE
You, 16/10

Setting or changing the alias gives a new link. The old link, printed QR codes and Apple Wallet passes made from it show “No card found”. The Google Wallet pass updates itself, and a new card email goes out automatically.

SOURCE
You, 16/10

Staff can change their alias as often as they like. Each change ends the previous link, and My card asks them to confirm before saving.

SOURCE
You, 16/10

Clearing an alias, by the staff member or an admin, brings back the plain link without a name.

SOURCE
You, 16/10

Before saving, staff can preview their card at the new link. Nothing changes until they save: the new link doesn't work yet and the old one keeps working.

SOURCE
You, 16/10

Admins can clear an alias that's offensive or copies someone else's name. Every alias change is logged and emailed to manage@minmed.sg.

SOURCE
You, 16/10

ADMINS

Two roles: Super admin and Admin.

SOURCE
You, 16/10

Only a Super admin can reactivate (restore) a card, give it a new link code or delete it.

SOURCE
You, 16/10

Cards can be imported from a spreadsheet at launch.

SOURCE
You, 16/10

Admins see view counts. No monthly report emails.

SOURCE
You, 16/10

Every admin change is logged and emailed to manage@minmed.sg.

SOURCE
Spec 0.2 §1.1

LEAVERS AND RECORDS

HR tells an admin as part of offboarding. On the person's last day their card is deactivated and soft-deleted, and they can no longer sign in.

SOURCE
You, 16/10

Soft-deleted cards are kept but hidden. A Super admin can restore one or delete it permanently.

SOURCE
You, 16/10 (PDPA)

Soft-deleted cards are listed under a Deactivated tab on the Cards page, which only Super admins see.

SOURCE
You, 16/10

BOT PROTECTION

One IP address making more than 3 card opens or contact downloads within 1 minute must pass a CAPTCHA.

SOURCE
You, 16/10

After passing, that browser isn't asked again for 30 minutes.

SOURCE
You, 16/10

A busy card is checked: more than 20 opens of one card in 10 minutes means everyone opening it gets the CAPTCHA for the next 30 minutes. This replaces “every 5th visitor”.

SOURCE
You, 16/10; industry standards

A pass ends early after 20 opens in 5 minutes.

SOURCE
You, 16/10; industry standards

For the first 2 weeks the rules only record. Then they switch on.

SOURCE
You, 16/10; industry standards

3

Features

Everything in the first version. Anything not listed here isn't being built.

Public card

`me.minmed.sg`

- Card page: photo, name, designation, contacts and About
- Save contact, Email, Call and WhatsApp
- QR code when opened on a computer
- WhatsApp link preview
- No longer active, not found and temporary error pages
- CAPTCHA check against bots
- Hidden from search engines

Staff

`me.minmed.sg/my-card`

- Minmed Google sign-in, own card only
- My card: read-only details, a name or alias for the link, a preview before saving, QR code download
- Card email with link, QR code and Wallet options
- Google Wallet pass that updates itself
- Apple Wallet pass they make themselves
- Ask for changes by replying to the card email

Admin console

`me.minmed.sg/admin`

- Google sign-in, admin list only
- Cards: list, search, add, import, edit
- Card details: link, alias (clear it if needed), QR code, card image, views, history
- Deactivate (soft delete); Super admin also restores, re-codes and deletes permanently
- Activity log
- Admins and Settings, including the company logo (Super admin)
- Email to manage@minmed.sg on every change, alias changes included

4

Users

Visitor

SIGNS IN?

No

WHO THEY ARE

Anyone who has a card's link or scans its QR code.

WHAT THEY DO

Views the card, saves the contact, gets in touch. May be asked to pass a CAPTCHA.

Staff member (card holder)

SIGNS IN?

Yes, company Google account, to their own My card page only

WHO THEY ARE

Minmed staff with a card.

WHAT THEY DO

Gets their card by email, keeps it in Wallet or on their phone, shares it. Can add a name or alias to their link and download their QR code. Asks an admin for other changes.

Admin

SIGNS IN?

Yes, company Google account

WHO THEY ARE

For example HR, added by a Super admin.

WHAT THEY DO

Adds, edits, sends and deactivates cards.

Super admin

SIGNS IN?

Yes, company Google account

WHO THEY ARE

One or two owners, for example manage@minmed.sg's owner and IT.

WHAT THEY DO

Everything an Admin does, plus admins, settings, and reactivating, re-coding or deleting cards.

manage@minmed.sg receives admin notifications automatically. It's a mailbox, not a user.

5

Roles and permissions

✓ allowed · — not allowed · **Own** their own card only · **Auto** the system does it. Every rule is checked on the server, not only by hiding buttons.

PUBLIC CARD

Open a card from its link or QR code

Visitor	Staff member	Admin	Super admin
✓	✓	✓	✓

Save contact, email, call, WhatsApp

Visitor	Staff member	Admin	Super admin
✓	✓	✓	✓

Forward the link to someone else

Visitor	Staff member	Admin	Super admin
✓	✓	✓	✓

Find a card without its link (search, browse, guess)

Visitor	Staff member	Admin	Super admin
—	—	—	—

GETTING AND KEEPING A CARD

Receive the card email (link, QR code, Wallet options)

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Add the card to Google Wallet or Apple Wallet

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Sign in to My card and see their own card details

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Set or change the name or alias in their own link

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Preview their card at a new link before saving

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Clear their own alias, going back to the plain link

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Download their own QR code

Visitor	Staff member	Admin	Super admin
—	Own	—	—

Change their own card details

Visitor	Staff member	Admin	Super admin
—	—	—	—

Ask for a change by replying to the card email

Visitor	Staff member	Admin	Super admin
—	Own	—	—

ADMIN CONSOLE

Sign in to the admin console

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

See and search all cards

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Add a card

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Import cards from a spreadsheet

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Edit a card's details and photo, and choose whether the mobile is shown

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Clear a staff member's alias

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Send or re-send the card email

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Download a card's QR code and card image

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Deactivate a card (soft delete)

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Restore a deactivated card

Visitor	Staff member	Admin	Super admin
—	—	—	✓

Issue a new link code (the old link stops working)

Visitor	Staff member	Admin	Super admin
—	—	—	✓

Delete a card permanently

Visitor	Staff member	Admin	Super admin
—	—	—	✓

See view counts

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

See the activity log

Visitor	Staff member	Admin	Super admin
—	—	✓	✓

Add or remove admins and change roles

Visitor	Staff member	Admin	Super admin
—	—	—	✓

Change the company logo shown on every card (in Settings)

Visitor	Staff member	Admin	Super admin
—	—	—	✓

Use a different logo on one card

Visitor	Staff member	Admin	Super admin
—	—	—	—

Change the brand colours, company details, notifications and Google Wallet settings

Visitor	Staff member	Admin	Super admin
—	—	—	✓

AUTOMATIC

Record every admin action and alias change in the activity log

Auto

Email manage@minmed.sg when a card is added, edited, given a new alias, deactivated, restored, re-coded or deleted

Auto

Send a new card email and update the Google Wallet pass when an alias changes

Auto

Update the Google Wallet pass when its card changes, and expire it when the card is deactivated

Auto

Ask for a CAPTCHA when a bot-protection rule is met (section 8)

Auto

6

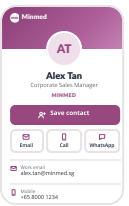
Site map and navigation

Four parts: the public card, the staff's My card page, the admin console, and the emails and passes that connect them. Select a page to see its mockup in section 7.

Public

`me.minmed.sg`

No sign-in · hidden from search engines · no menu



Card page

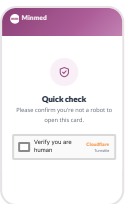
`/card/{code}`

or /card/{alias}/{code}



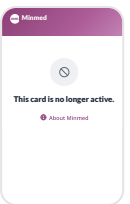
On a computer

adds a QR code



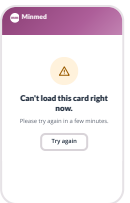
CAPTCHA check

when a rule in section 8 is met



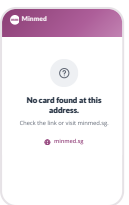
No longer active

after deactivation



Temporary error

card can't be checked



Card not found

any wrong address

Files behind the same code

`/functions/…?code={code}`

contact file, card image, Google Wallet link; served by backend functions

Home

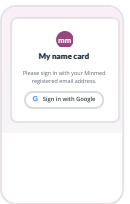
`/`

redirects to minmed.sg; no staff list

Staff

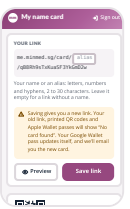
`me.minmed.sg/my-card`

Minmed Google sign-in · own card only



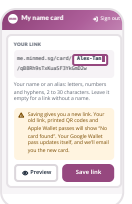
Sign in

`/my-card`



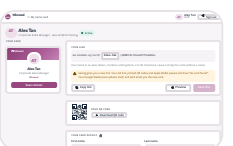
My card

every field read-only, one box for the link, QR code download



Changing the link

type, preview, confirm, done



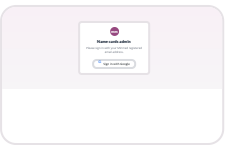
On a computer

same page, wider

Admin console

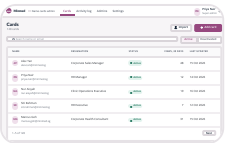
`me.minmed.sg/admin`

Company Google sign-in · admin list only



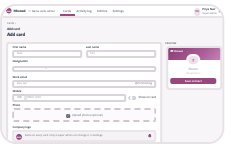
Sign in

`/admin`



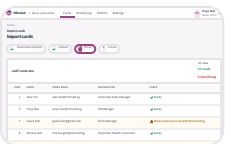
Cards

`/admin/cards`



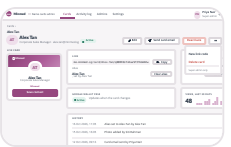
Add card

`/admin/cards/new`



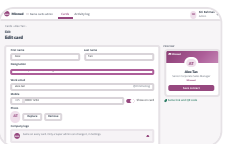
Import cards

`/admin/cards/import`



Card details

`/admin/cards/{id}`

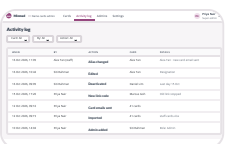


Edit card

`/admin/cards/{id}/edit`

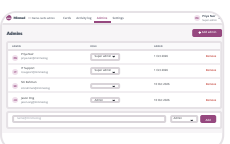
Confirm dialogs

deactivate, clear alias, new link code, delete



Activity log

`/admin/activity`



Admins

`/admin/admins`

Super admin only



Settings

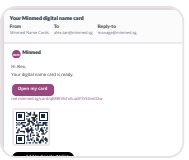
`/admin/settings`

Super admin only

Emails, Wallet and sharing

`outside the website`

How cards reach staff, admins and visitors



Card email

to the staff member



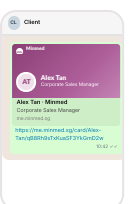
Google Wallet pass

issued by Minmed



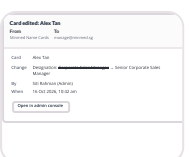
Apple Wallet pass

made by the staff member



WhatsApp link preview

when a link is shared

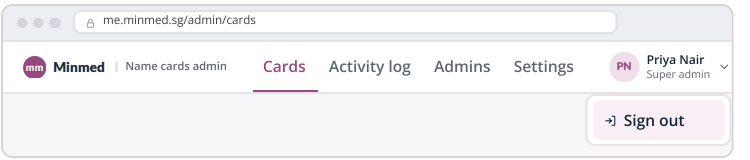


Admin notification

to manage@minmed.sg

Figure 2. Site map. Each entry is a page or message, with its address.

SUPER ADMIN



ADMIN

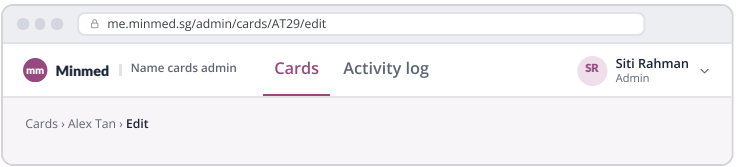


Figure 3. Admin console navigation.
Admins see Cards and Activity log; Super admins also see Admins and Settings. The account menu holds Sign out. Inside a card, the breadcrumb leads back to Cards.

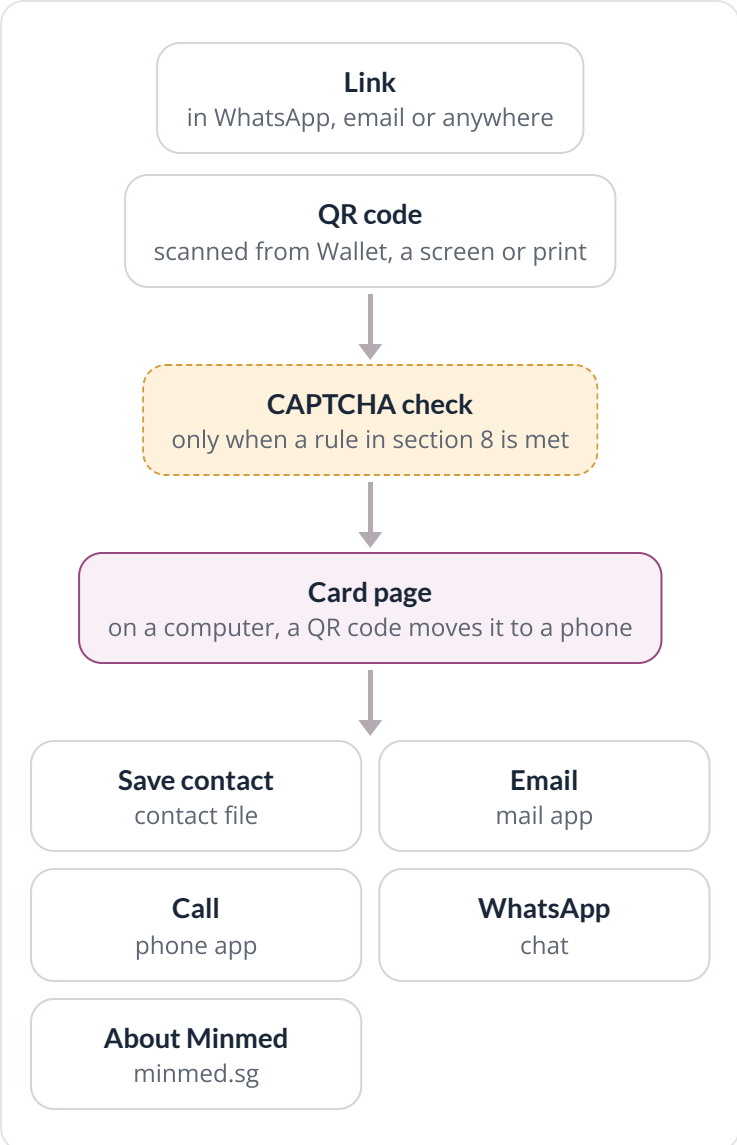


Figure 4. How a visitor moves through the public card. There's no menu and no way to reach another person's card. Call and WhatsApp appear only when a mobile is published.

7

Pages

High-fidelity mockups of every page and message. You approved the wording on 16/10/2026, except the preview screen's (open item 2). The “mm” logo stands in for the real one, and the QR codes can't be scanned.

Public card

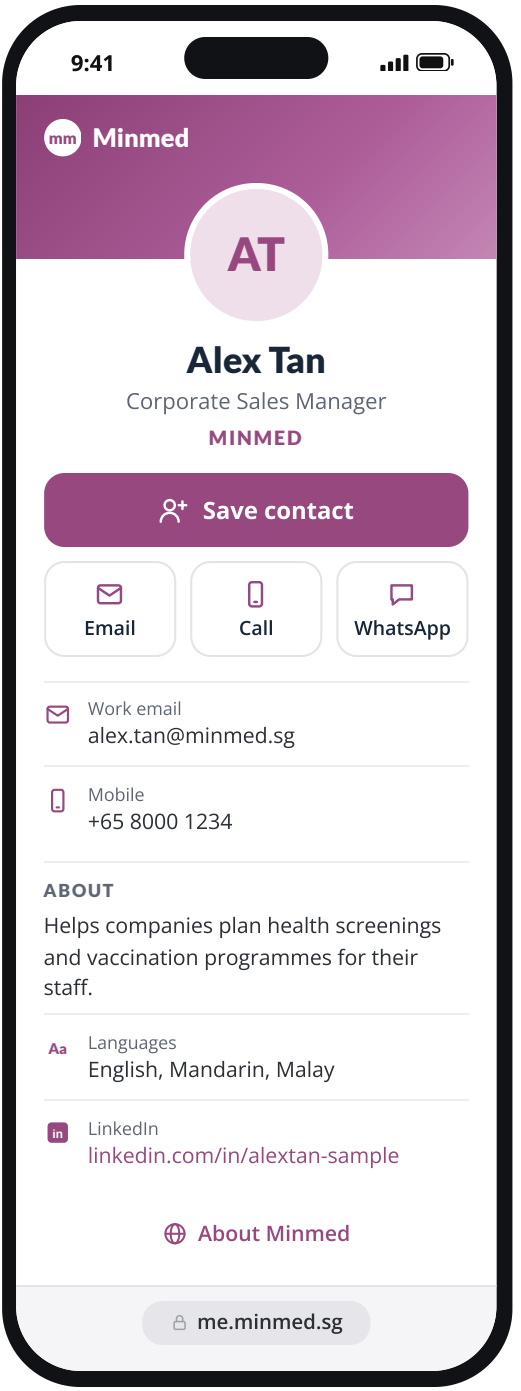


Figure 5. Card page.

`me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`, or `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w` until a name or alias is added · anyone with the link. Call and WhatsApp only with a published mobile; About only when filled in.

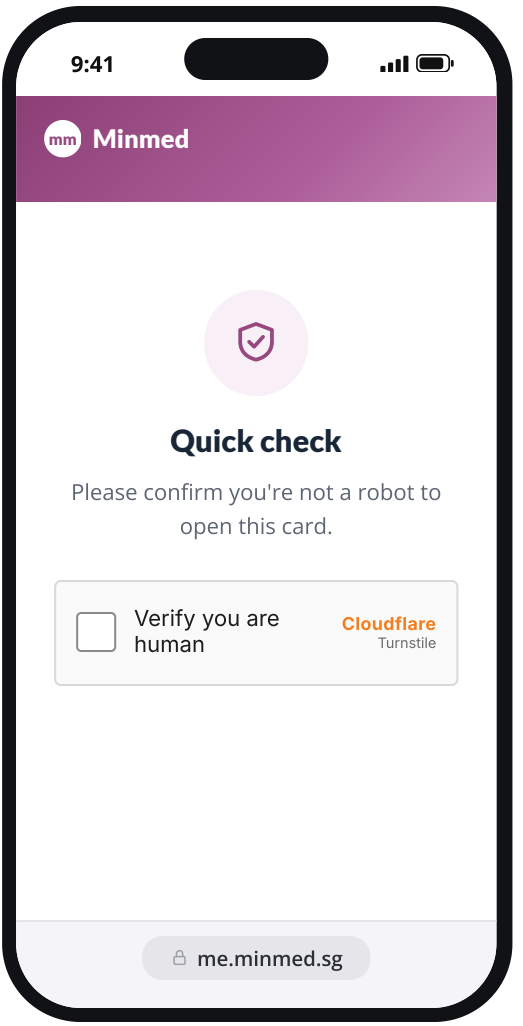


Figure 6. CAPTCHA check.
Same address, shown instead of the card when a rule in section 8 is met. Most people pass without clicking; this is what a flagged visitor sees.

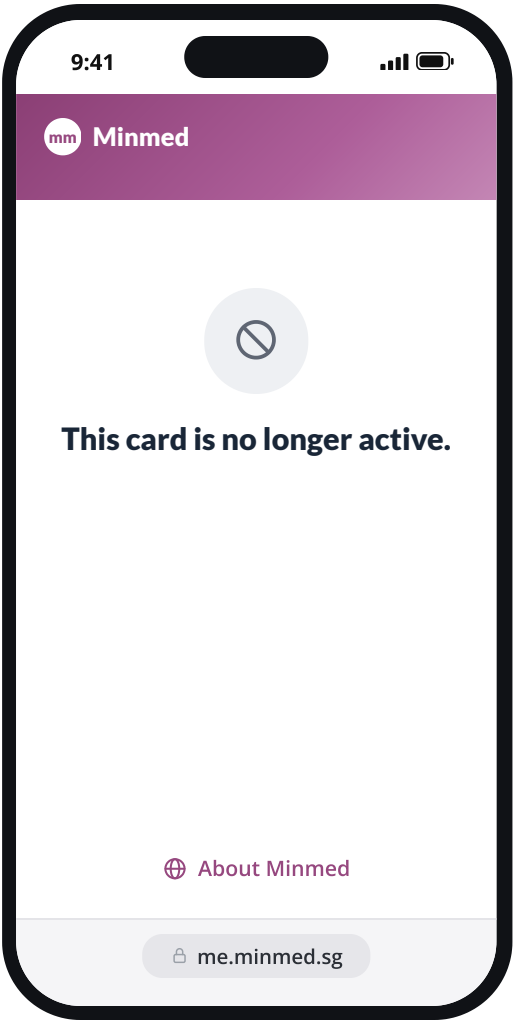


Figure 7. No longer active.
Same address after deactivation. No details, photo, Save contact or contact buttons.

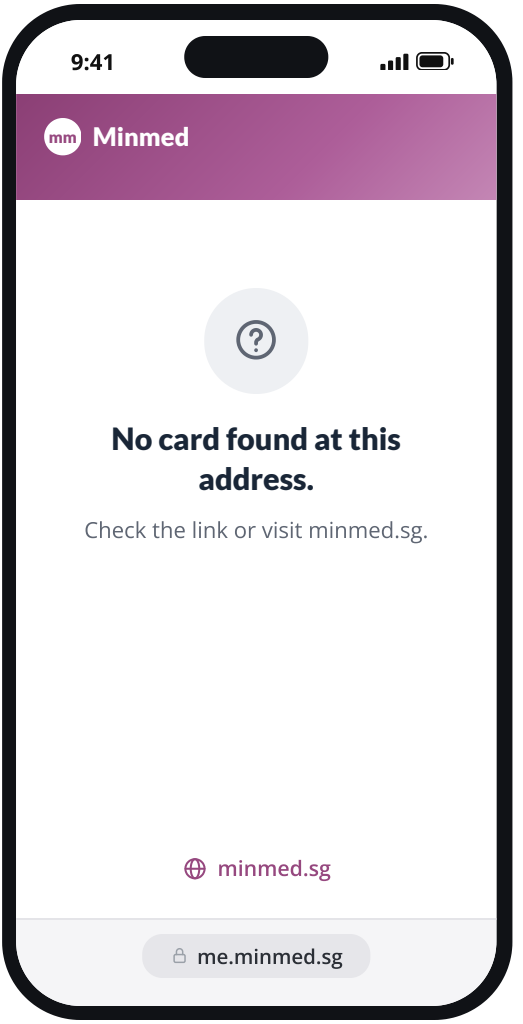


Figure 8. Card not found.
Any wrong code, old alias, deleted card or mistyped link. The same answer every time, so nobody learns which codes exist.

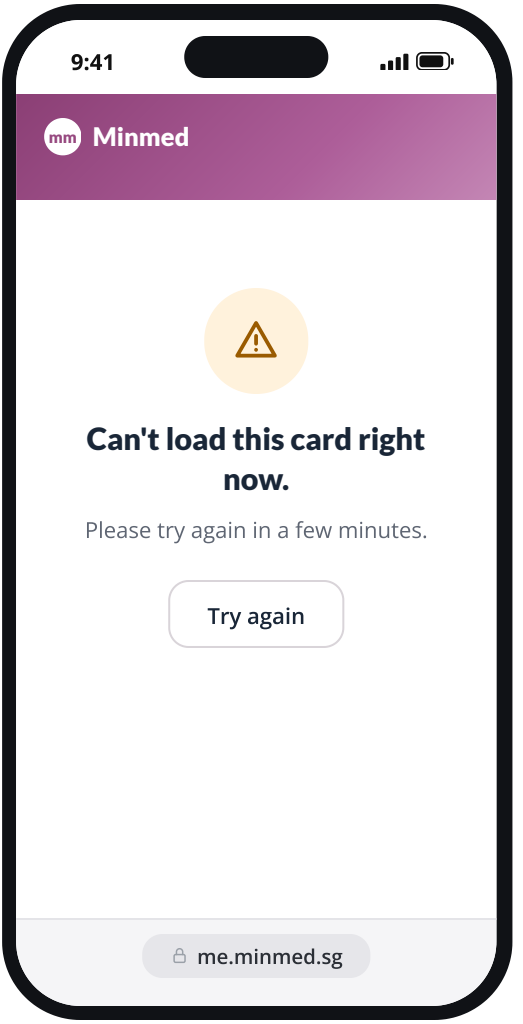


Figure 9. Temporary error.
The card can't be checked, for example during an outage or when Base44 credits run out. Never shows old details.

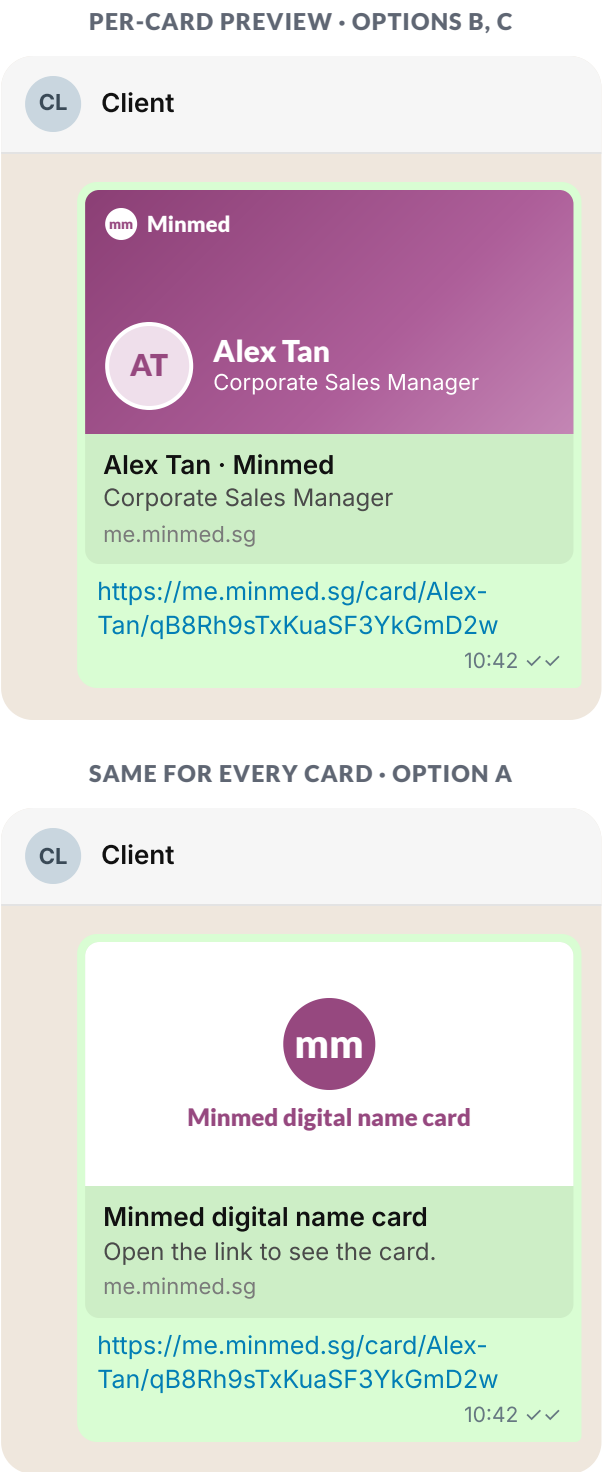


Figure 10. WhatsApp link preview. TEST

Which one depends on the WhatsApp test (open item 1).

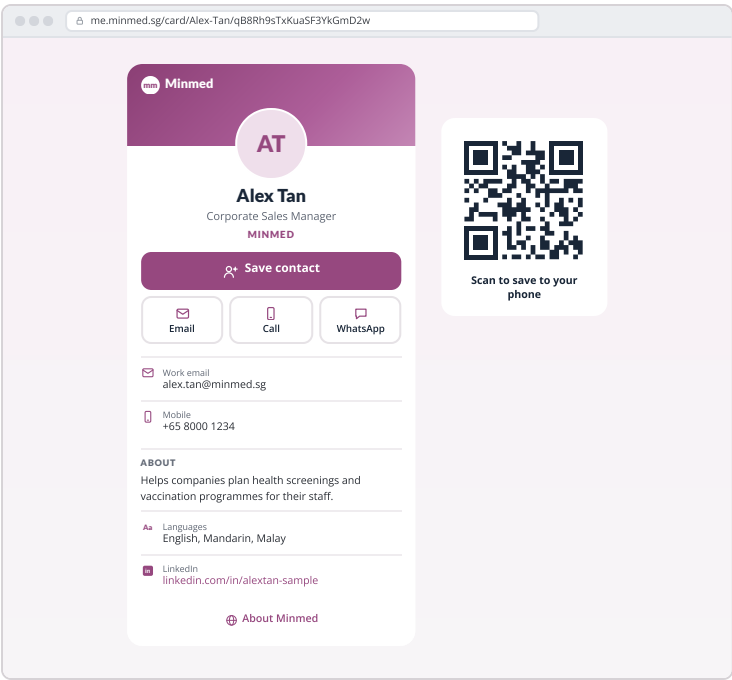


Figure 11. Card page on a computer.
The same card in a narrow column, plus a QR code to move it to a phone.

Your Minmed digital name card

From Minmed Name Cards **To** alex.tan@minmed.sg
Reply-to manage@minmed.sg

Minmed

Hi Alex,

Your digital name card is ready.

Open my card

me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w

Add to Google Wallet

iPhone (iOS 27)

1. Take a screenshot of the QR code.
2. In Wallet, tap + then Create a Pass.
3. Choose the screenshot.

No Apple Intelligence? Choose Create Pass Manually and scan the QR code from a computer screen.

Want your name in your link? Sign in at me.minmed.sg/my-card.

Need another change? Reply to this email.

Figure 12. Card email to the staff member.
Sent when an admin chooses Save and send card email or Send card email, and again after every alias change. Replies go to manage@minmed.sg.

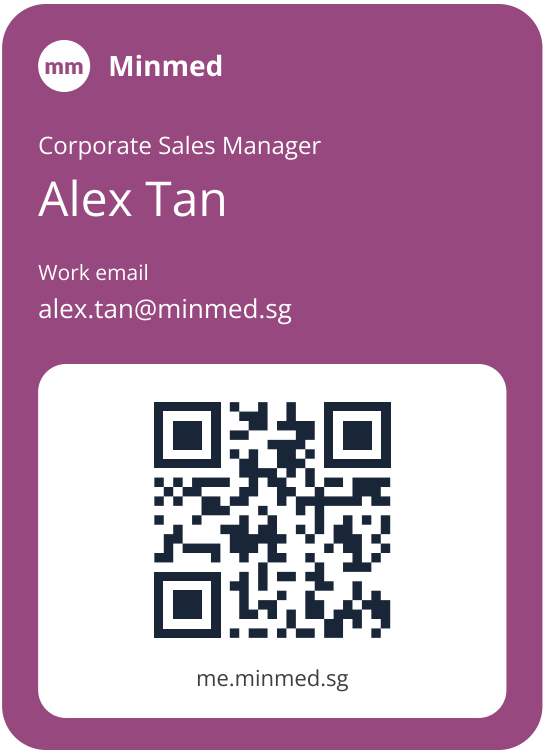


Figure 13. Google Wallet pass.
Issued by Minmed. Updates when the card changes and shows as expired when it's deactivated.

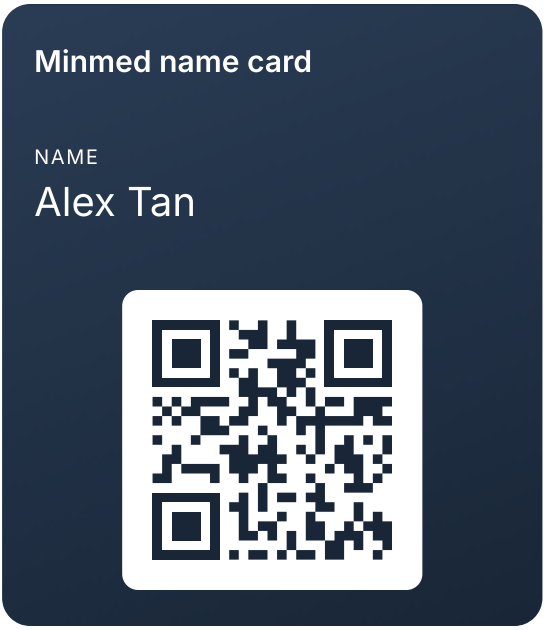


Figure 14. Apple Wallet pass, made by the staff member.

Made with iOS 27's Create a Pass, so its look may vary. Minmed can't update or remove it, but its QR code always opens the live card.

My card (staff view)

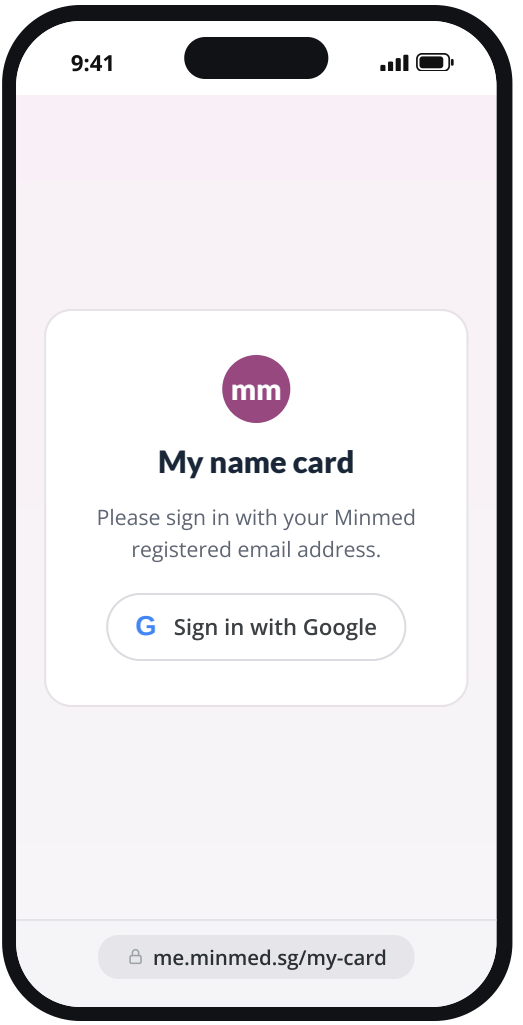


Figure 15. My card sign-in.

`me.minmed.sg/my-card`. A Minmed account without an active card sees “There's no name card for this account.”

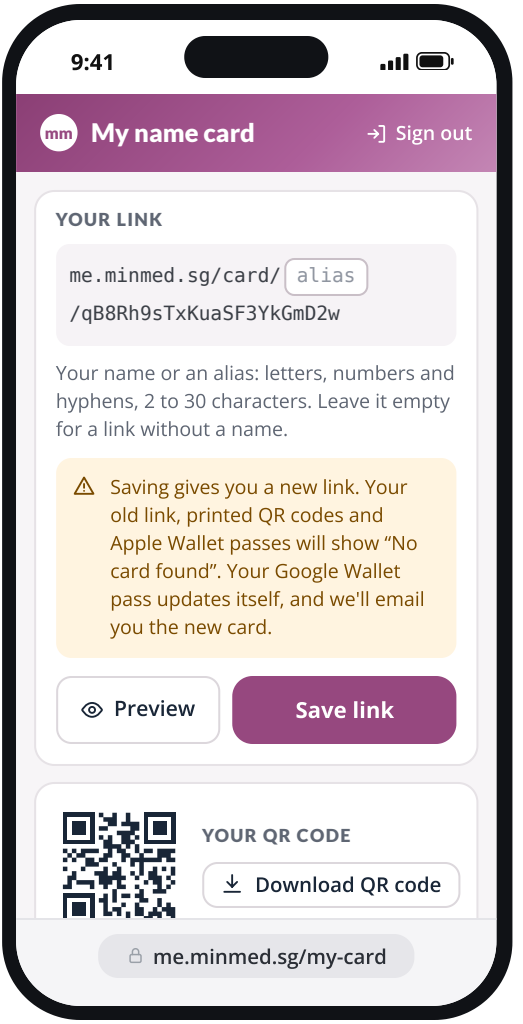
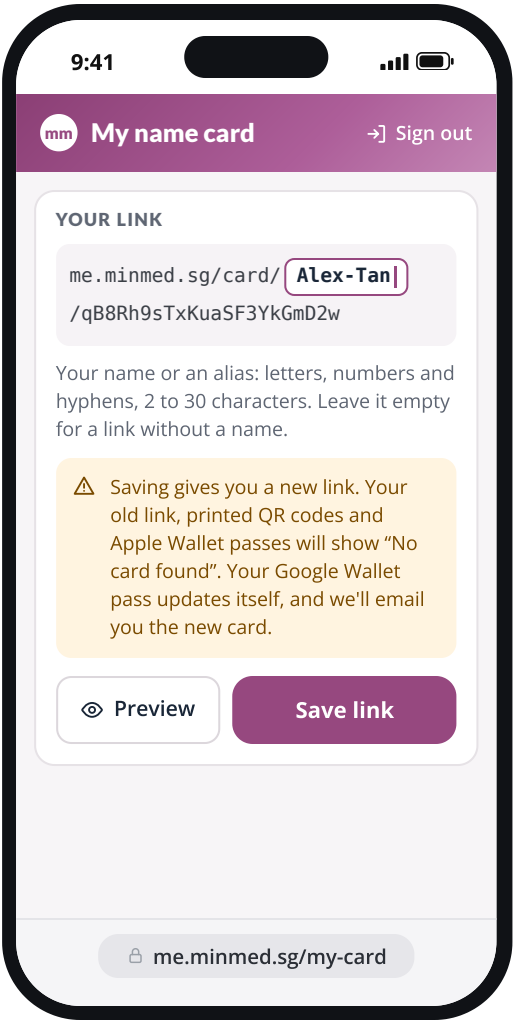


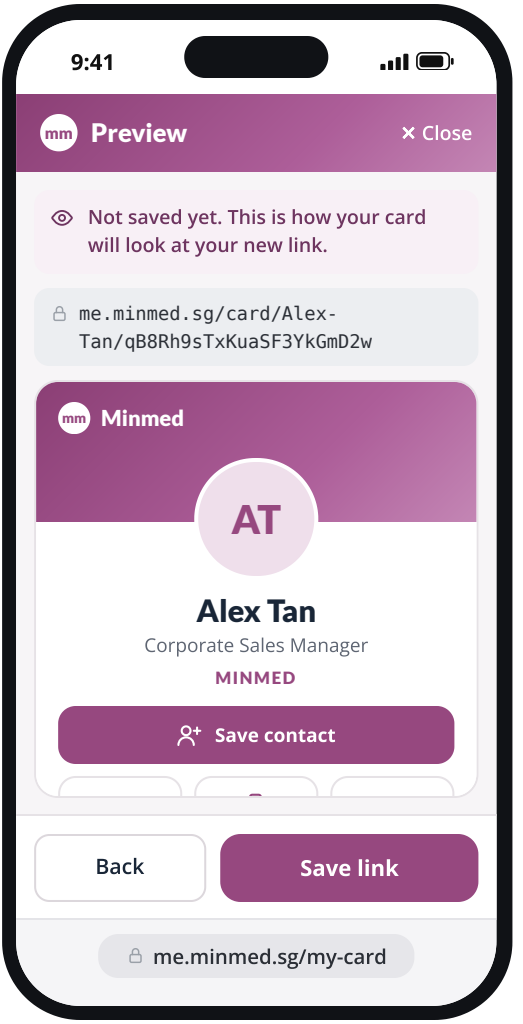
Figure 16. My card.

Top of the page after signing in. The only box staff can type in is the name or alias in their link. Preview shows the card at the new link before saving, and they can download their QR code. Below that, every field is read-only, including the company logo (whole page in Figure 18).

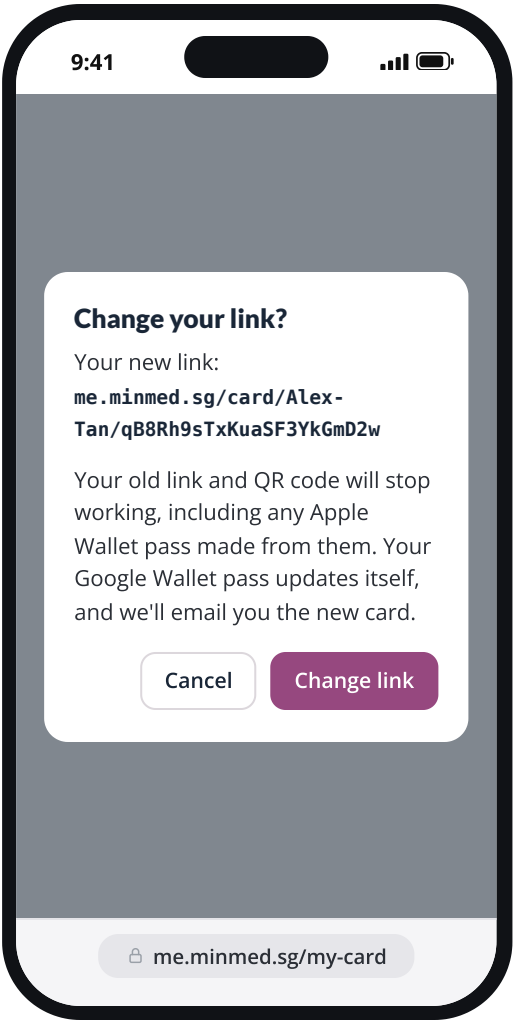
1 Type a name or alias



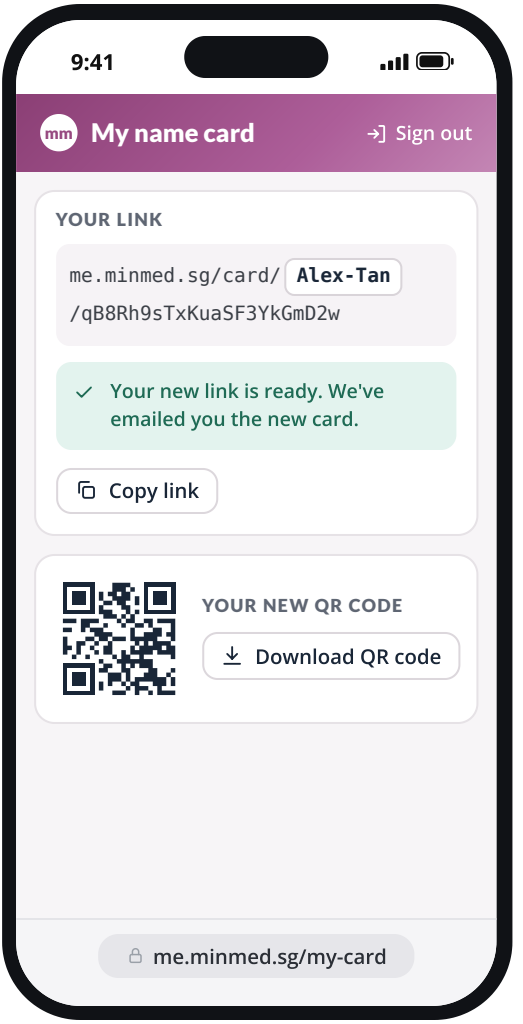
2 Preview the card before saving



3 Save and confirm the new link



4 Done: new link, QR code and email



5 If the alias breaks the rule

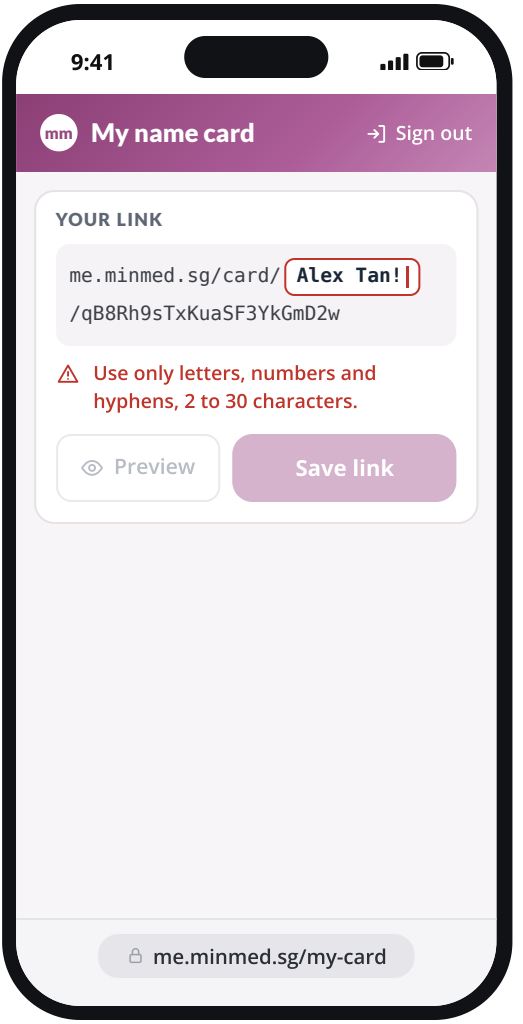


Figure 17. How a staff member changes their link.

1. They type a name or alias in the box. 2. Preview shows their card at the new link; nothing is saved yet. 3. They save and confirm. 4. The new link works at once and a new card email goes out; the old link shows “No card found”. 5. An alias that breaks the rule can't be previewed or saved.

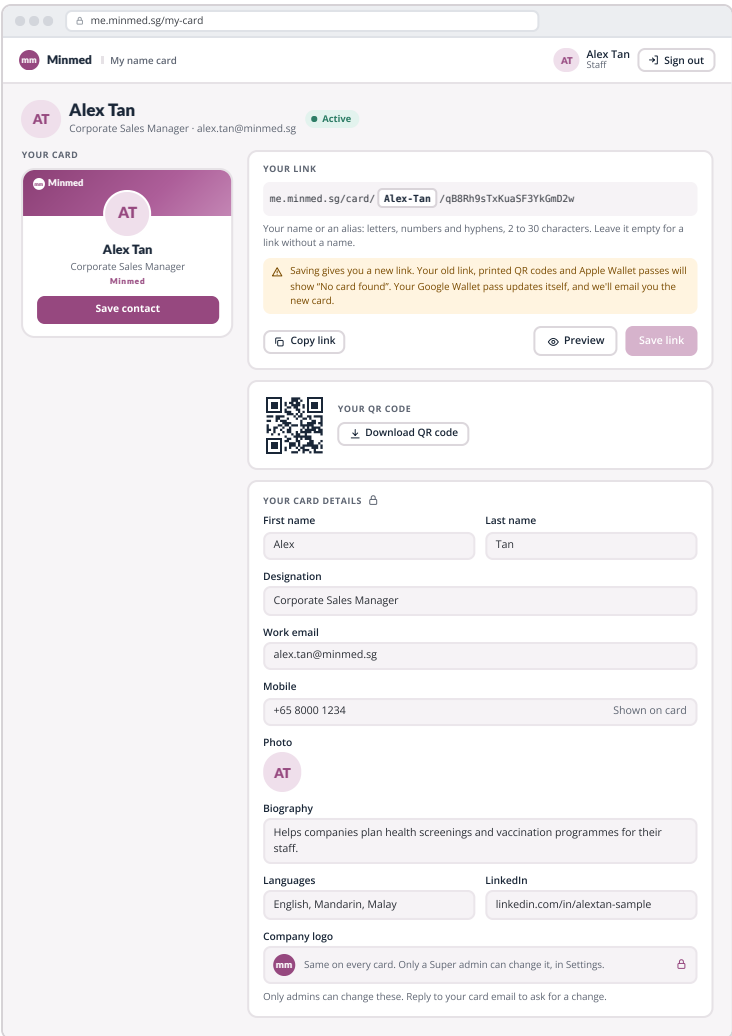
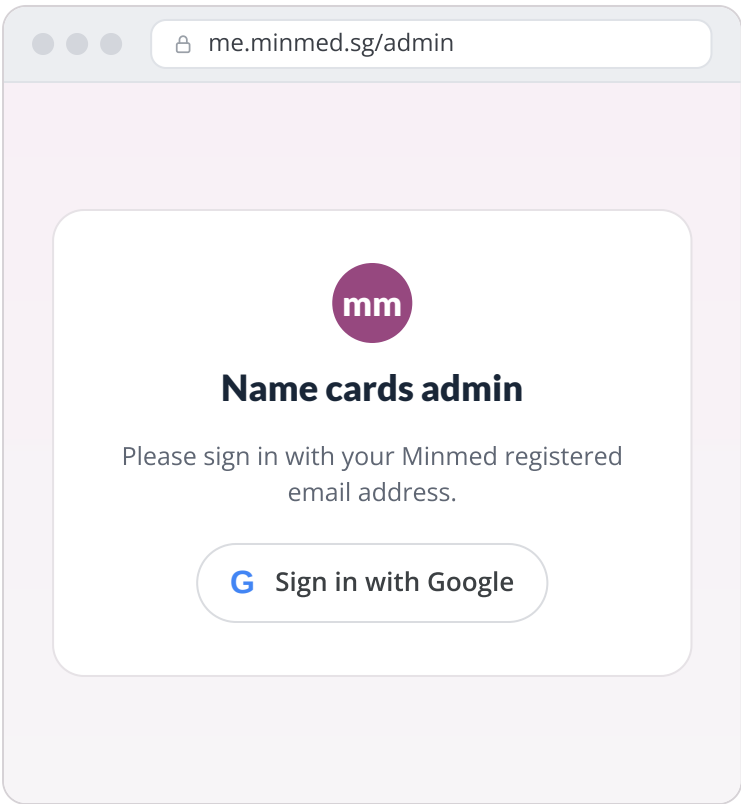


Figure 18. My card on a computer.

The same page, laid out like the admin card details page: the card on the left, then the link box, the QR code and every field read-only.

Admin console



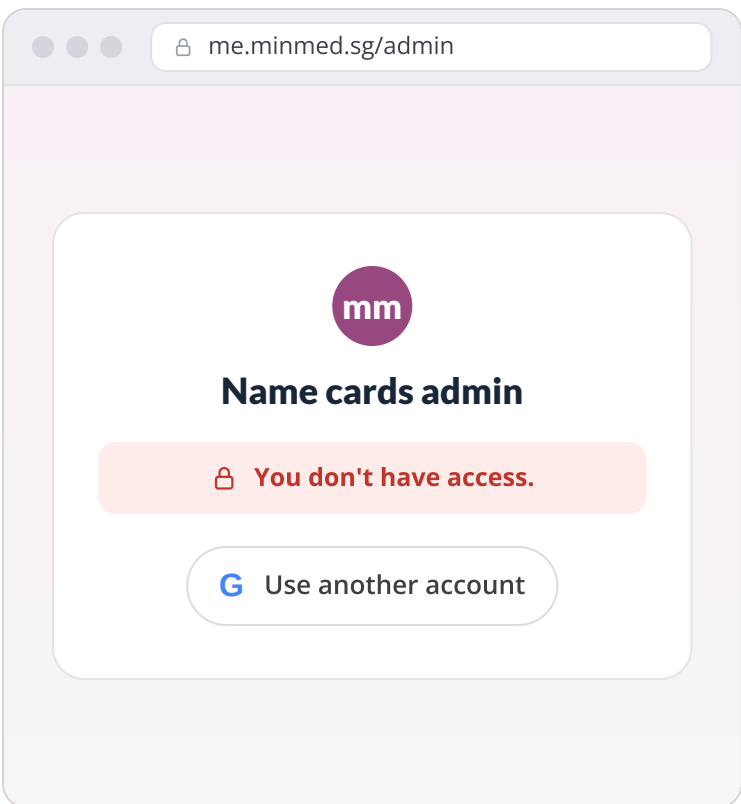


Figure 19. Admin sign-in.

Left: `me.minmed.sg/admin`. Right: a company account that isn't on the admin list.

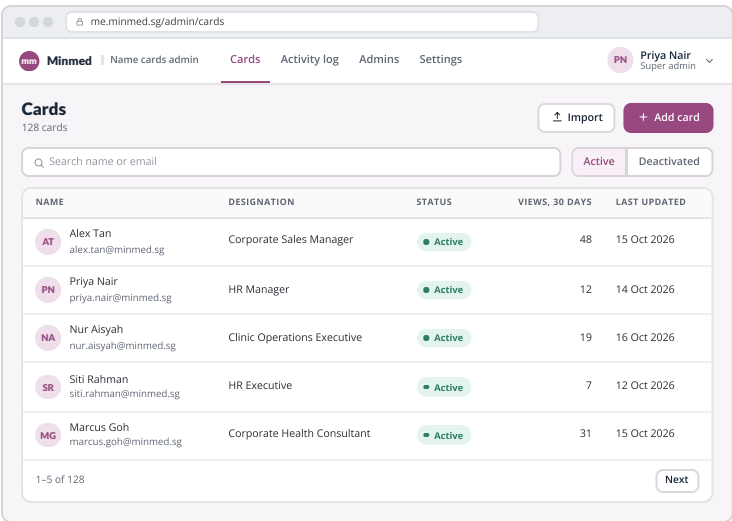


Figure 20. Cards.

`/admin/cards`. Search by name or email. Deactivated cards are soft-deleted: they leave this list and appear under Deactivated, for Super admins. Views count card page openings in the last 30 days.

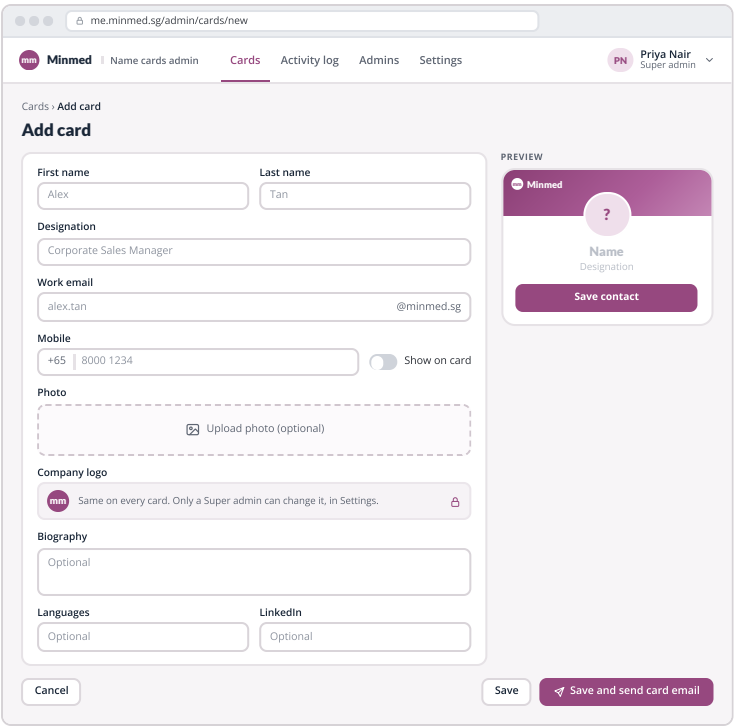


Figure 21. Add card.

`/admin/cards/new`. Work email must end with @minmed.sg. The mobile shows on the card only when switched on. The company logo is locked: it's the same on every card.

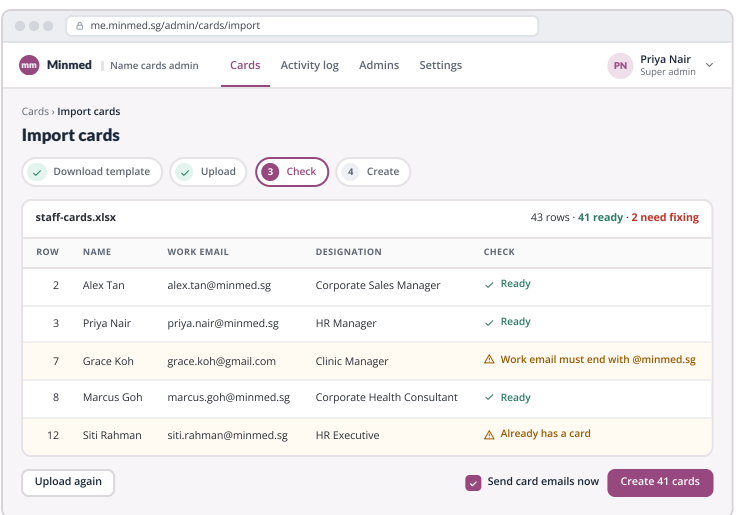


Figure 22. Import cards.

`/admin/cards/import`. Rows with problems are skipped until fixed. The card emails can go out in the same step.

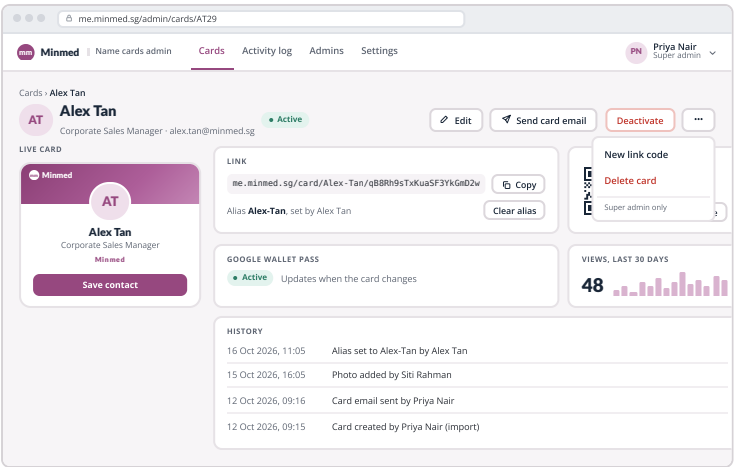


Figure 23. Card details.

`/admin/cards/{id}`. Admins can clear an alias. The ⋯ menu (New link code, Delete card) appears only for Super admins. On a deactivated card, Restore replaces Deactivate.

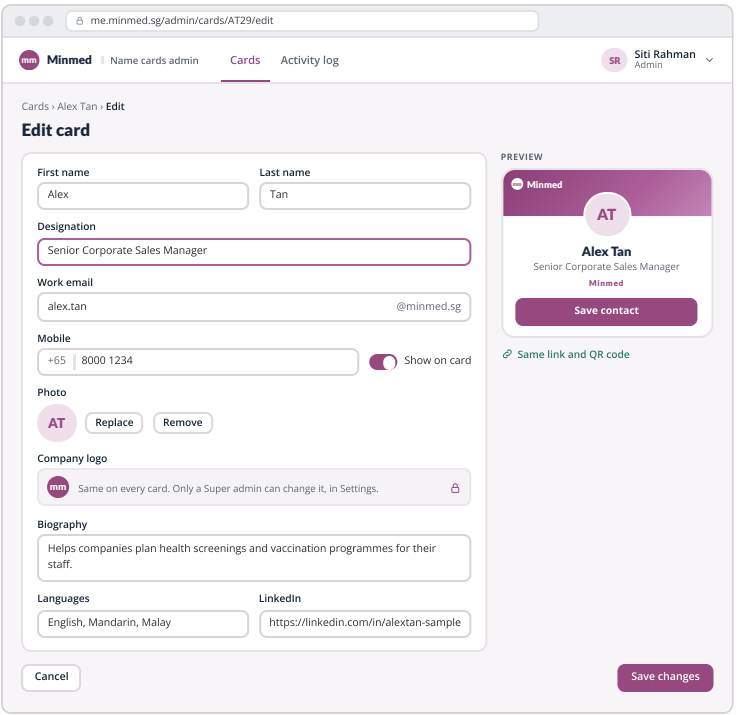


Figure 24. Edit card.

`/admin/cards/{id}/edit`, seen by an Admin, who has no Settings tab. Saving changes the live card at once, on the same link and QR code, and emails manage@minmed.sg. The company logo is locked here for everyone.

Deactivate Alex Tan's card?

The link will show “This card is no longer active.” Save contact stops working and Alex can't sign in. The card is soft-deleted: it leaves the Cards list, and a Super admin can restore it.

Cancel | Deactivate

Clear Alex Tan's alias?

The link goes back to me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w. The link with the alias stops working, and Alex gets a new card email.

Cancel | Clear alias

Give Alex Tan's card a new link?

The old link and QR code will stop working.

☑ Send Alex the new card email

Cancel | New link code

Delete Alex Tan's card permanently?

This can't be undone. The link will show “No card found”.

Cancel | Delete permanently

Figure 25. Confirm dialogs.
Deactivate and Clear alias are for Admins and Super admins. New link code and Delete are for Super admins only.

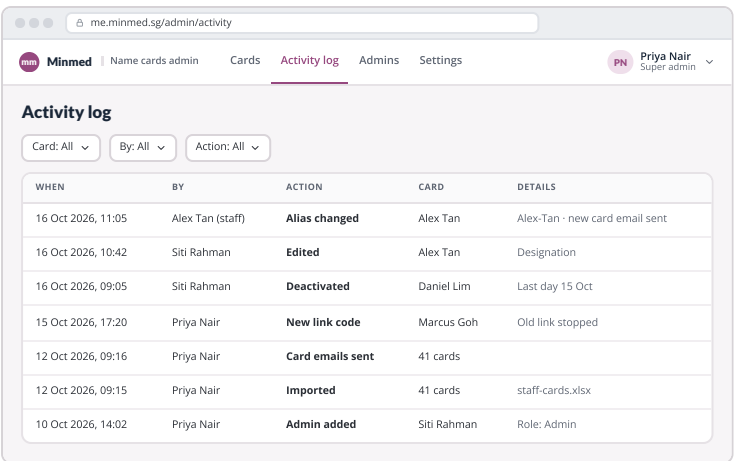


Figure 26. Activity log.
`/admin/activity`. Every admin action and alias change: who, what, which card and when.

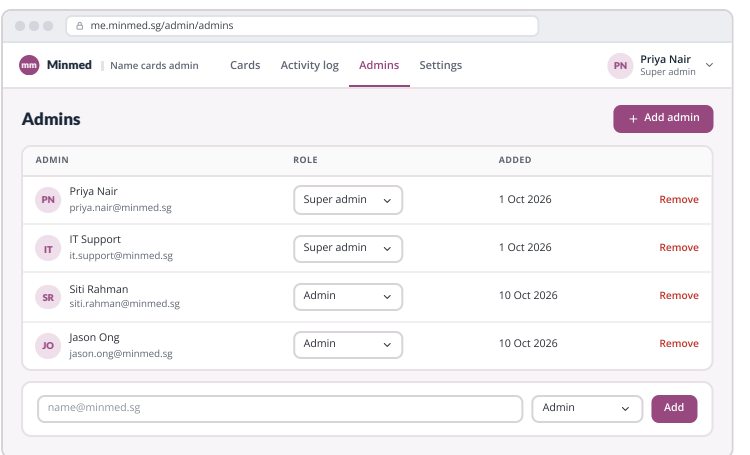


Figure 27. Admins (Super admin only).
`/admin/admins`. Admins are added by company email.

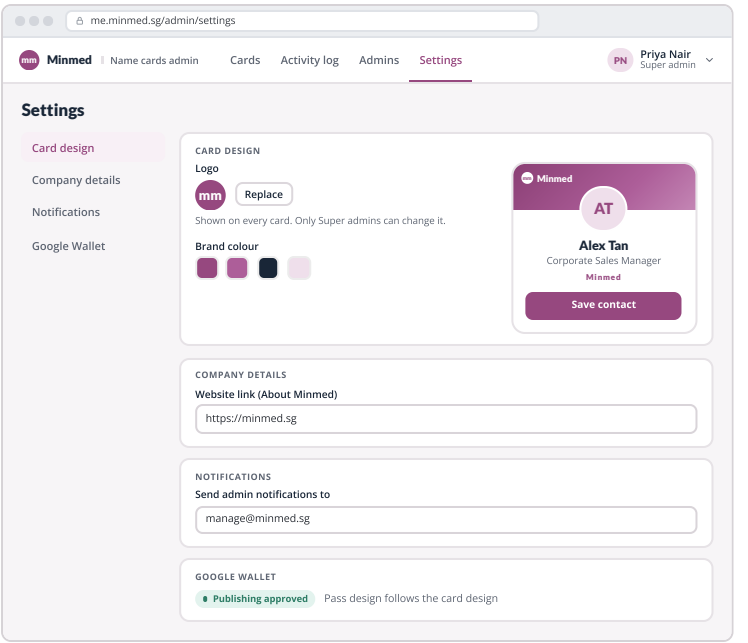


Figure 28. Settings (Super admin only).

`/admin/settings`. Card design: the company logo on every card, which only Super admins can change. Also company details, notifications and Google Wallet.

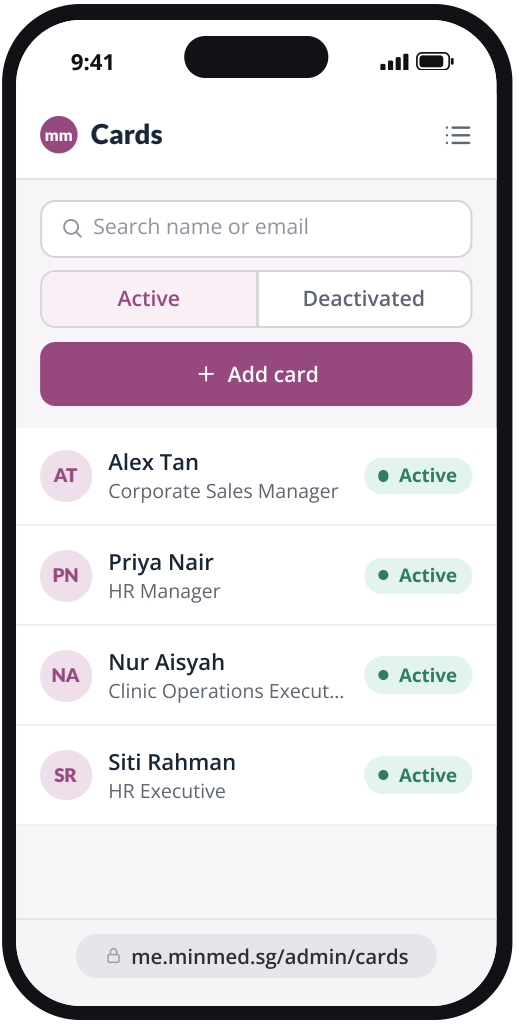


Figure 29. Admin console on a phone.

The same Cards page on a phone: the table becomes a list, with the same Active and Deactivated tabs.

Card edited: Alex Tan

From Minmed Name Cards **To** manage@minmed.sg

Card	Alex Tan
Change	Designation: ~~Corporate Sales Manager~~ → Senior Corporate Sales Manager
By	Siti Rahman (Admin)
When	16 Oct 2026, 10:42 am

Open in admin console

Alias changed: Alex Tan

From Minmed Name Cards **To** manage@minmed.sg

Card	Alex Tan
Change	Link: ~~me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w~~ → me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w
By	Alex Tan (staff)
When	16 Oct 2026, 11:05 am

Open in admin console

Figure 30. Admin notification.
Sent to manage@minmed.sg for every add, edit, alias change, deactivation, restore, new link code and deletion. Two examples: an admin's edit and a staff member's alias change.

8

Bot protection (CAPTCHA)

Purpose: slow down bots that copy cards in bulk. Random links and hiding from search stop people finding cards; these rules limit how fast anyone can open them. A CAPTCHA can't stop a person, or a bot that pays a solving service, from copying a card they can already open.

Rule 1

Same IP

WHAT'S COUNTED

Card opens, contact downloads and wrong card addresses from one IP address

LIMIT

More than 3 in a rolling minute

THEN

CAPTCHA before anything is sent

STATUS

CONFIRMED

Rule 2

Pass

WHAT'S COUNTED

A browser that has passed the CAPTCHA

LIMIT

30 minutes

THEN

No new CAPTCHA for that browser

STATUS

CONFIRMED

Rule 3

Busy card

WHAT'S COUNTED

Opens and downloads of one card, from all IP addresses together

LIMIT

More than 20 in a rolling 10 minutes

THEN

Everyone opening that card gets the CAPTCHA for the next 30 minutes

STATUS

CONFIRMED

Replaces “every 5th visitor”

Rule 4

Limit after a pass

WHAT'S COUNTED

Opens and downloads by a browser holding a pass

LIMIT

More than 20 in 5 minutes

THEN

The pass ends and the CAPTCHA comes back

STATUS

CONFIRMED

Never checked

WhatsApp and other link-preview fetches (they only see the preview), the admin console and My card (Google sign-in), and images, styles and scripts.

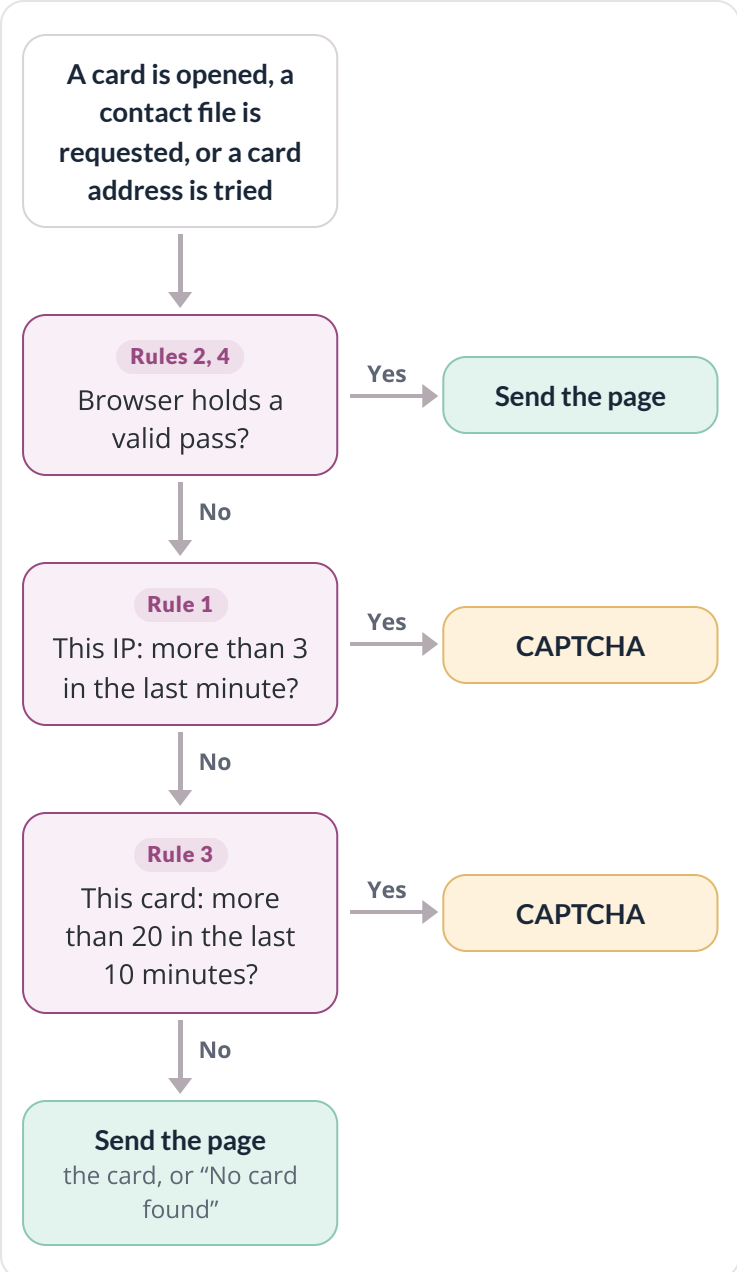


Figure 31. How each request is checked.
Counting happens on the server, in rolling windows.

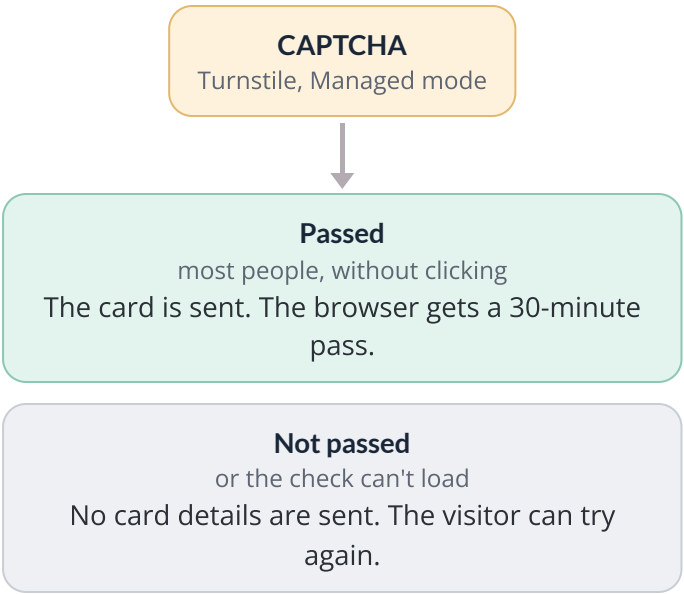


Figure 32. What happens at the CAPTCHA.
The page is in Figure 6. A Turnstile token is checked by a backend function; it works once and expires after 5 minutes.

Based on industry standards and research

Each rule follows published guidance from OWASP (a web security nonprofit), Cloudflare and Google.

Use a CAPTCHA only as a step-up for suspicious traffic, not as the main defence. None of the sources suggests checking every Nth visitor: that would interrupt 1 in 5 real people, while a bot just tries again.

SOURCE
OWASP

USED IN
Rules 1 and 3, instead of “every 5th visitor”

Limit by more than the IP address. For example, count all requests to one page, whoever sends them.

SOURCE

OWASP, Cloudflare

USED IN

Rule 3

Count in rolling windows, not fixed ones, so bursts can't hide at the edges.

SOURCE

OWASP

USED IN

All rules

A pass lasts 30 minutes by default; 15 to 45 minutes is recommended.

SOURCE

Cloudflare

USED IN

Rule 2

Keep limits in place after a CAPTCHA is passed.

SOURCE

OWASP

USED IN

Rule 4

Against scraping, use two limits per IP: 10 requests in 2 minutes, then 20 in 5 minutes.

SOURCE

Cloudflare

USED IN

Rule 4's limit

Prefer managed checks that only ask a person to click when needed.

SOURCE
Cloudflare

USED IN
Turnstile in Managed mode

Run the checks without acting first, then set the limits from real traffic.

SOURCE
Google, Cloudflare

USED IN
The first 2 weeks record only

The numbers are starting points. Rule 4's limit matches Cloudflare's example; rule 3's is our estimate for a name card. Both get checked during the record-only weeks.

- **Managed mode:** Turnstile decides whether to show a checkbox. Most people never see it, and there are no picture puzzles.
- **Shared networks:** office Wi-Fi, events and mobile data put many people behind one IP address, so they meet rule 1 sooner. So does one person who opens a card, saves it and reloads twice. In Managed mode, most still pass without clicking.
- **Server-side:** the card page asks a backend function for the card. When a check is needed, no card details are sent until Cloudflare confirms the token.
- **Launch** CONFIRMED: for the first 2 weeks the rules only record. Compare them with real traffic, then switch them on.

Sources: OWASP bot management, Cloudflare rate limiting best practices, Cloudflare challenge passage, Cloudflare

challenge types, Google reCAPTCHA v3, Turnstile widget modes and server-side validation.

9

Links and privacy

- **Link format:** `me.minmed.sg/card/{code}`, or `me.minmed.sg/card/{alias}/{code}` once the staff member adds a name or alias. For example `me.minmed.sg/card/qB8Rh9sTxKuaSF3YkGmD2w` and `me.minmed.sg/card/Alex-Tan/qB8Rh9sTxKuaSF3YkGmD2w`. The code has no name, initials or counting numbers. The `/card/` prefix keeps clear of Base44's own addresses.
- **The code** CONFIRMED: 22 random letters and numbers, made by the server with a secure random generator (CSPRNG) and never reused. That's about 131 bits of randomness, above the 128-bit minimum in the standard below.
- **The alias:** optional and chosen by the staff member: letters, numbers and hyphens, 2 to 30 characters. It adds no security; the code alone keeps the link unguessable. A link with an alias can be found by searching for that alias.
- **Both parts are checked:** the code must belong to a card, and the alias must be the card's current one, or absent when none is set. Anything else shows “No card found” and counts towards rule 1.
- **The link changes only** when the alias is set, changed or cleared, or a Super admin issues a new code. Name and designation edits keep the same link.

- **An alias change ends the old link:** links already sent, printed QR codes and Apple Wallet passes made from it show “No card found”. The Google Wallet pass updates itself and a new card email goes out.
- **Every public address uses the same random code:** the card page, contact file, card image and Google Wallet pass. Never a counting number, so nobody can fetch cards by counting.
- **Hidden from Google, twice over:** no list of cards anywhere (no sitemap entries, no staff directory, the home address redirects to minmed.sg), and every card page, image and contact file tells search engines not to index it. On Base44 that means the page's “Search visibility” switch plus a noindex header on files from backend functions, checked after every publish. Don't block card addresses in robots.txt: Google then can't see the “don't index” instruction and may still list a link someone posted elsewhere.
- **If a link is misused:** a Super admin issues a new code. The old link then shows “No card found”. An admin can also clear an offensive alias.
- **Bots:** see section 8.

Based on industry standards and research

A link that is the only key to a page must be unguessable: a random identifier, never a counting number.

SOURCE
W3C

USED IN
Random code

Random strings meant to be unguessable need a secure random generator and at least 128 bits of randomness. Each bit doubles the number of possible codes.

SOURCE
OWASP ASVS 5.0

USED IN
22 characters

A fixed or predictable part, such as a name or initials, adds no strength.

SOURCE
OWASP

USED IN
The alias adds none; the code alone keeps links unguessable

Researchers scanned every possible 5- and 6-character short-link code. They found shared OneDrive files and Google Maps directions that revealed people's home addresses.

SOURCE
Georgiev and Shmatikov, 2016

USED IN
Long codes; rule 1 counts wrong addresses

Sources: W3C Good Practices for Capability URLs (2014 draft), OWASP ASVS 5.0 requirement 11.5.1, OWASP session management and Gone in Six Characters: Short URLs Considered Harmful for Cloud Services.

10

Feasibility on Base44 and Wallet

Checked against Base44, Apple, Google and Cloudflare documentation on 16/10/2026. Base44's settings are new and its docs contradict each other in places, so anything marked TEST needs trying on a throwaway app before the build starts. The WhatsApp preview options and the tests are in the build notes, a separate file.

Custom domain me.minmed.sg

STATUS

WORKS

WHAT IT MEANS FOR THE SCOPE

One DNS record pointing to Base44, on the Starter plan or higher. If minmed.sg's DNS is on Cloudflare, the record must be “DNS only”, so Cloudflare's own protection can't sit in front.

Card address with optional alias

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Base44 keeps addresses at the root for sign-in and system pages, so card links need a prefix: `/card/{code}` and `/card/{alias}/{code}`. Two-part addresses aren't documented.

Hidden from Google

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

A per-page “Search visibility” switch (added 28/09/2026) marks pages noindex and removes them from the sitemap. Files served by backend functions can carry their own noindex header. Check what the live site serves after every publish.

Per-card WhatsApp preview

STATUS

NOT SUPPORTED AS IS

WHAT IT MEANS FOR THE SCOPE

Base44 sets preview tags per page template, not per card, so every card would share one Minmed preview. The options and plan are in the build notes.

CAPTCHA (Cloudflare Turnstile)

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Free plan: unlimited checks, up to 20 widgets, and no need to route traffic through Cloudflare. The page loads Turnstile (Cloudflare suggests a React component); a backend function verifies each token using a secret key kept in Base44's secrets. Base44's security policy can't be changed per app, so check that Turnstile loads.

Visitor IP address

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Rule 1 needs the visitor's real IP address inside a backend function. Base44 passes the request headers on, but doesn't document which one holds the visitor's IP. Log the headers on a test app. If none does, rule 1 needs option C's separate host.

Counting for the CAPTCHA rules

STATUS

OURS TO BUILD

WHAT IT MEANS FOR THE SCOPE

Cloudflare can't sit in front of a Base44 domain, so the rules are counted in backend functions with a database table, in rolling windows.

Contact file and Google Wallet link

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Backend functions can return files of the right type and use code libraries. Their addresses look like `me.minmed.sg/functions/contact?code=…`, still behind the random code.

Sign-in limited to minmed.sg (admins and staff)

STATUS

WORKS, WITH SETUP

WHAT IT MEANS FOR THE SCOPE

Needs Minmed's own Google sign-in set to “Internal” (Builder plan or higher). Base44 only has “admin” and “user” roles, so Super admin and Admin are our own list, editable only by Super admins. Staff sign in as ordinary users. Base44's plans don't charge per user.

Staff limited to their own card and alias

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Base44's row-level rules can limit a signed-in user to the card with their email, and field-level rules can limit which fields they write. Saving goes through a backend function that checks the alias, then updates the link, the Google Wallet pass, the activity log and the emails. Check that the field rule can be tied to the person's own card.

Soft delete for leavers

STATUS

WORKS

WHAT IT MEANS FOR THE SCOPE

Base44 records carry a soft-delete flag and date, so a deactivated card can be hidden but kept, and restored by a Super admin.

Visitors can't list cards

STATUS

OURS TO BUILD

WHAT IT MEANS FOR THE SCOPE

A public Base44 app lets anyone register. Visitors must only get one card's public fields, by its code, through a backend function. One wrong rule could expose every card.

Emails to staff and manage@minmed.sg

STATUS

WORKS, WITH SETUP

WHAT IT MEANS FOR THE SCOPE

Needs a paid plan and a verified sending domain. One recipient per email, about 1–2 credits each.

View counts

STATUS

WORKS

WHAT IT MEANS FOR THE SCOPE

Record card openings ourselves; Base44's own analytics keeps 60 days. A view is a card page opening, not a person. Link previews and admin previews don't count.

Rate limits and credits

STATUS

BUILT IN

WHAT IT MEANS FOR THE SCOPE

Base44 rate-limits public endpoints by default; the limits aren't published or adjustable. Function calls, counting and emails use credits. If credits run out, those actions pause and visitors see the temporary error page. Watch usage.

Google Wallet

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

Free issuer account, but Google must approve publishing (needs a business profile). Passes can be updated and set to expired.

Apple Wallet

STATUS

TEST

WHAT IT MEANS FOR THE SCOPE

iOS 27's Create a Pass: from a screenshot of the QR code on iPhone 15 Pro or newer with Apple Intelligence; on other iOS 27 iPhones (iPhone 11 or newer), Create Pass Manually and scan the QR code. No Apple fee. Minmed can't update or remove these passes.

Sources: Base44 docs on external domains, search visibility, SEO and sharing, backend functions, data security, sending emails, security, troubleshooting and credits; Apple's Create passes in Wallet; Google's Generic pass, publishing access and expired passes; Cloudflare's Turnstile plans and Turnstile overview.

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Open items

The only open items. Everything else is confirmed.

1 **WhatsApp previews: option A, B or C.**

TEST

Plan confirmed: test B first. Then C if per-card previews are a must, otherwise A. The options are described in the build notes. Affects Figure 10, section 10 and the file addresses in section 6. Needs the test result.

2 **Wording on the preview screen.**

PROPOSED

Step 2 of Figure 17: the title Preview, the note “Not saved yet. This is how your card will look at your new link.”, the new link shown above the card, and Back and Save link buttons. The Preview button sits next to Save link (Figure 16, Figure 18). Needs your OK.

Sources. Minmed Digital Name Cards Technical Specification draft 0.2 (15/10/2026). Your decisions in this conversation on 16/10/2026. Standards, research, platform and vendor documentation linked in sections 8, 9 and 10.